

LC 983 - Min Cost For Tickets

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Targeted by: Flipkart, Zepto, Intuit

1. Problem Understanding

We are given 2 vectors: `days` and `costs` .

- **days:** The days at which we'll travel.
- **costs:** The cost for passes. It contains:
 - 1 day pass cost
 - 7 day pass cost
 - 30 day pass cost

2. Example Walkthrough

Let's consider an example:

`days = [1, 4, 6, 7, 8, 20]`

`costs = [2, 7, 15]`

Method	Strategy	Calculation	Total Cost
M1	Buy 1 day tickets for all travel days	6 days $\times$ 2\$	12 \$
M2	Buy 7 day pass at day 1 $\rightarrow$ covers (1, 4, 6, 7) Now for (8, 20) $\rightarrow$ buy 1 day pass each	7\$ (2\$ + 2\$) = 4\$	11 \$ (Best Option)

3. Intuition & Approach

Since the minimum cost depends on the ticket bought, I'll use **Dynamic Programming**.

```
dp[i] = min cost to cover the first i travel days
```

State Transitions:

Now at $dp[i]$, we have 3 choices:

- **1 day ticket:** $cost = dp[i] + costs[0]$
- **7 day ticket:** $cost = dp[i] + costs[1]; (Covers\ until\ days[j] < days[i] + 7)$
- **30 day ticket:** $cost = dp[i] + costs[2]; (Covers\ until\ days[j] < days[i] + 30)$

```
dp[j] = min(dp[j], cost)
```

4. C++ Solution Implementation

Below is the structured code reflecting our DP choices:

```

class Solution {
public:
    int mincostTickets(vector<int>& days, vector<int>& costs) {

        int n = days.size();
        vector<int> dp(n + 1, INT_MAX);

        // dp will store minimum cost to travel i days
        dp[0] = 0;

        for(int i = 0; i < n; i++) {
            dp[i + 1] = min(dp[i + 1], dp[i] + costs[0]); // 1 day ticket

            int j = i;
            while(j < n && days[j] < days[i] + 7) { // 7 din ka ticket liya to 7 din tak m
                j++;
            }
            dp[j] = min(dp[j], dp[i] + costs[1]);

            j = i;
            while(j < n && days[j] < days[i] + 30) { // 30 din ka ticket liya to 30 din ta
                j++;
            }
            dp[j] = min(dp[j], dp[i] + costs[2]);
        }

        return dp[n];
    }
};

```